

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I M. Madhulika (192472188) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Madhulika.

QUESTIONS:4

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

cloud Based video conferencing platform.

1.1 Analyze whether the problem is primarily caused by TCP/UDP or the application layer.

The problem is primarily related to the transport protocol, especially the use of TCP instead of UDP.

- video conferencing is a real-time application that requires low latency and continuous data delivery.
- TCP provides reliable delivery using acknowledgement, retransmission, sequencing and congestion control.
- During peak hours, packet loss and congestion can cause TCP to retransmit packets.
- These transmissions increase latency and jitter, resulting in delayed audio/video.
- UDP does not retransmit lost packets, so it can provide faster real time communication.
- However, the application layer also has an important role in adapting video quality according to the available bandwidth.

conclusion: The major issue is likely a transport-layer protocol mismatch, such as using TCP for real time media, combined with application-level adaptation problems.

1.2 Justify the answer based on communication requirements of real time multimedia application.

- 1) Low latency - audio and video must arrive quickly.
- 2) Low jitter - packets should arrive at relative consistent interval.
- 3) High throughput - sufficient bandwidth is required for video.
- 4) Loss tolerance - a small amount of packet loss is preferable to long delay.
- 5) Continuous transmission - delays packets may become useless.
- 6) Adaptive quality - video resolution and bitrate should adjust according to network conditions.

TCP focuses on reliable and ordered delivery, while real-time communication generally prioritizes timeliness over perfect reliability.

Therefore, UDP-based protocols are generally more suitable for real time audio/video.

2) Multinational software company - DNS Resolution

2.1 Analyze the reasons behind the high DNS resolution time.

- users are geographically far from DNS servers
- DNS queries may travel through multiple ~~server~~ servers

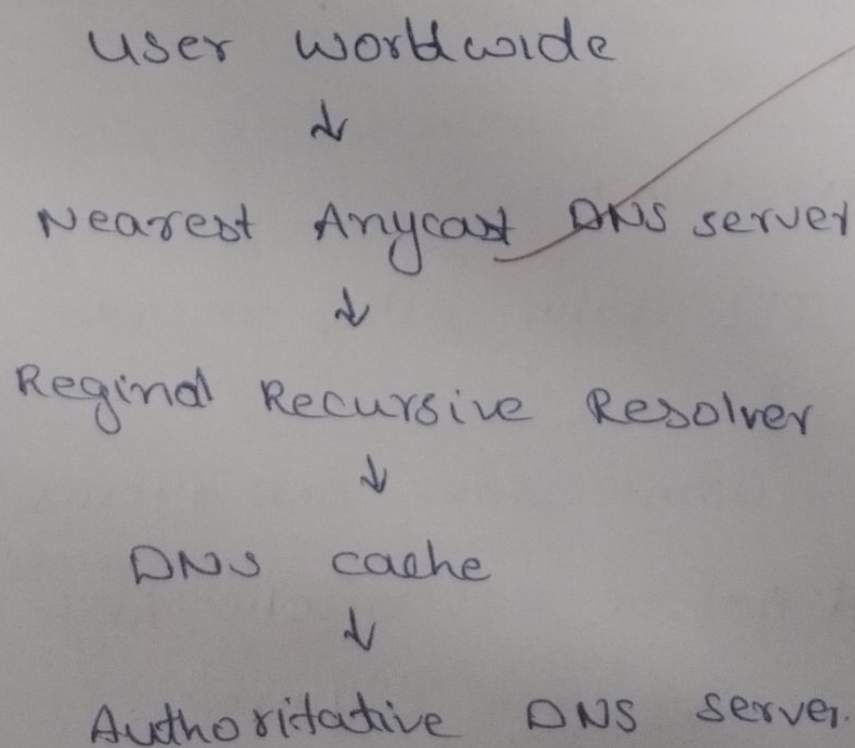
- poorly configured or overloaded DNS servers
- Low DNS cache hit ratio
 - very short TTL values causing frequent DNS lookups
 - Slow authoritative DNS servers.
 - Network congestion and Packet loss.
 - Slow authoritative Multiple DNS records requiring additional lookups
 - Lack of geographically distributed DNS infrastructure

A centralized DNS architecture can also create a single point of failure and performance bottle-neck.

2.2 Propose a suitable DNS architecture.

A suitable architecture is a distributed Any-cast based DNS architecture

Architecture



Techniques used :-

1) Anycast DNS.

- The same IP address is advertised from multiple geographic locations.
- users are automatically routed to a nearby DNS server.
- This reduces network latency.

2) DNS prefetching.

- DNS resolving can be performed before the user actually requests a resource.
- This request reduces perceived page loading time.

3) DNS caching.

- Frequently requested DNS records are stored in recursive resolvers.
- cached responses avoid repeated queries to authoritative servers.

4) TTL optimization :-

- Appropriate TTL value allows records to remain cached longer.
- This reduces unnecessary DNS queries.
- TTL should not be excessively high when rapid DNS changes are required.

cloud service provider - Global DNS.

3.1 Impact of centralized DNS architecture on performance.

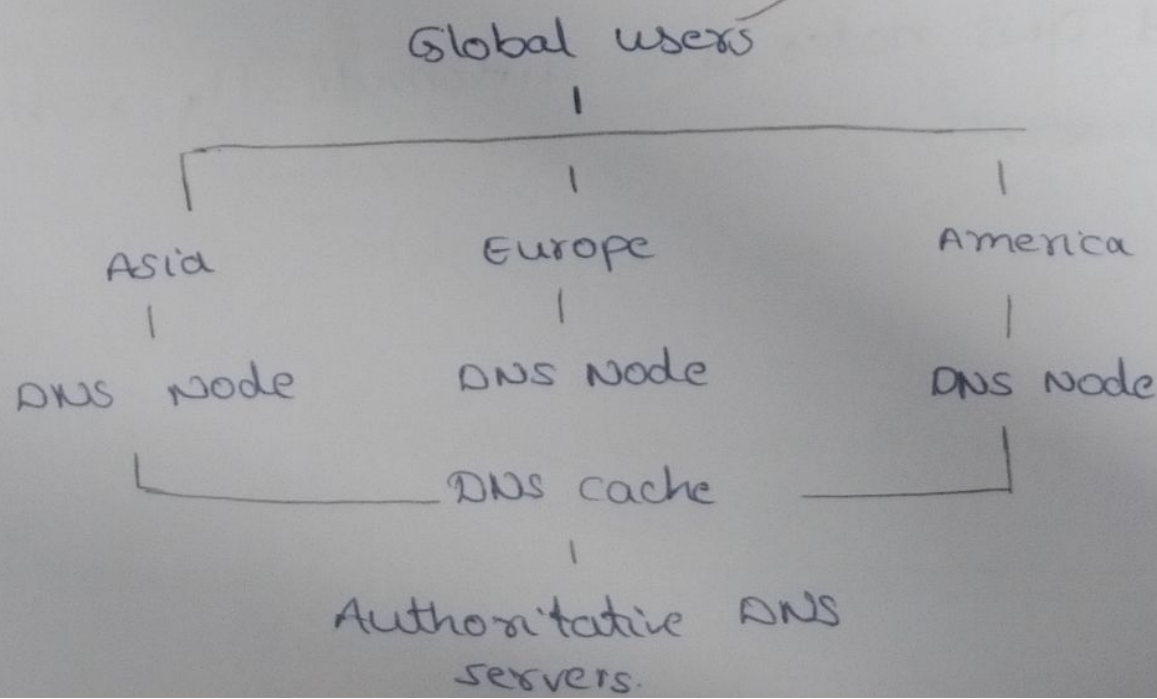
A centralized DNS architecture can become a performance bottleneck when millions of users generate DNS queries.

Problem include:-

- Long distance DNS requests.
- Increased network latency.
- High query processing load.
- DNS server overload.
- single point of failure.
- Poor scalability during traffic spikes.
- Increased response time during promotional events.

3.2 Distributed DNS Solution.

A globally distributed Anycast DNS architecture should be implemented.



Main components:

1) Anycast routing.

- Multiple DNS servers use the same IP address.
- BGP routers use towards the nearest available DNS location.

2) DNS caching

- Frequently requested domain records are cached at regional resolvers.
- Reduces repeated authoritative queries.

3) Multiple authoritative DNS servers

- DNS Data is replicated across several locations.
- Failure of one server does not stop DNS resolution.

4) Load balancing.

- Queries are distributed among DNS nodes.
- Prevents individual servers from becoming overloaded.

5) Health Monitoring.

- Failed DNS nodes are automatically removed from service.

Financial stock Trading Platform.

Three possible TCP-related factors responsible for latency.

The major factors are.

1) TCP Retransmission.

Packet losses causes TCP to retransmit packets.

Packet loss $\rightarrow$ Retransmission $\rightarrow$ waiting $\rightarrow$ increased latency

In a high frequency trading system, even a small delay can affect transaction execution

2) TCP congestion control.

TCP continuously adjusts its sending rate according to network conditions

During congestion:

- TCP reduces its congestion window
- Transmission speed decreases
- Packets may be delayed.
- Transaction latency increases

3) Nagle's Algorithm.

Nagle's Algorithm combines small TCP packets into larger packets to improve network efficiency.

However, in stock trading application may send many small, time sensitive messages

Monitor TCP performance.

- RTT
- Packet loss
- Retransmission rate.
- congestion window.
- connection establishment time.

conclusion

For a high frequency Platform, TCP-NODELAY, Persistent connections, optimized buffers, reduced packet loss and low latency network paths can significantly improve transaction speed and reliability. The system should carefully balance reliability with the extremely low latency required by financial trading.

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